

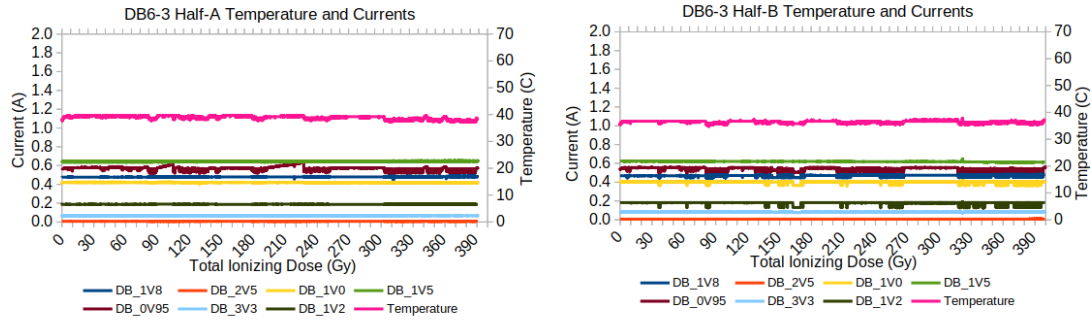


Figure 3. TID test of DB6-3. All board currents and the temperatures of both KU FPGAs were monitored over the whole irradiation period delivered by a 54 MeV proton beam.

to the IS25LP128F FLASH memories, where after annealing some of the devices recovered the re-programing capabilities, with all the contents of the memory not being affected in any of the devices.

Four different variants of 10 Gbps multi-mode SFP+ devices were tested: AVAGO, CORETEK, FS-Europe and Direktronik. The two AVAGO AFBR-709SMZ devices completely failed to pass the irradiation tests done up to 9×10^{12} (1 MeV equivalent n)/cm² fluence. One out of nine CORETEK CT-A000NPP-SB1L-D devices tested failed to function after irradiating to 9×10^{12} (1 MeV equivalent n)/cm² fluence. In the case of the Direktronik 33-4248 and the FS-Europe SFP-10GSR-85 devices, one of each of the two brands was exposed to a higher fluence of 14×10^{12} (1 MeV equivalent n)/cm², and a batch of ten of each of the two brands was exposed to 7×10^{12} (1 MeV equivalent n)/cm² fluence. All of the samples irradiated from Direktronik and FS-Europe were fully functional after irradiation.

It was seen that the threshold voltages for the FDFMA3N109 MOSFETs used in the design shifted with the exposure (Figure 4) due to the Gamma component present in the reactor cavity (195 Gy). The asymmetrical profile of the flux in the cavity was used to have a perception of the correlation between the shifts of the thresholds in the MOSFETs and the exposed fluence. Four out of fifteen

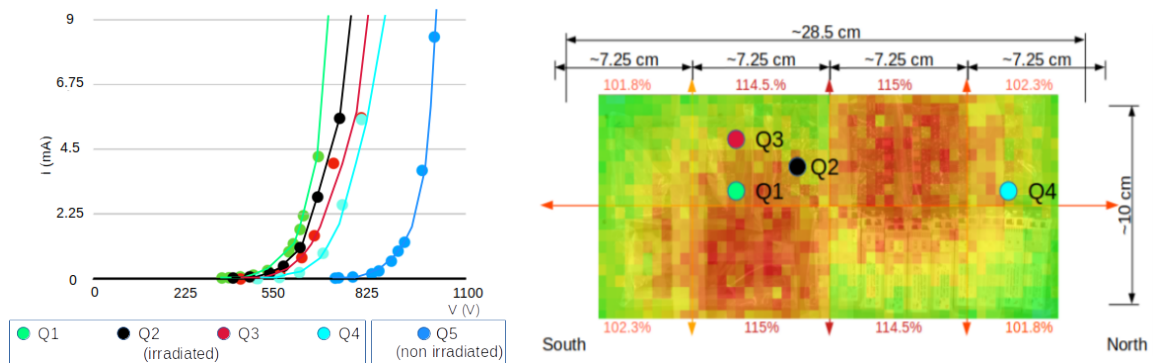


Figure 4. (Left:) Threshold shift depicted in an IV curve of four of the irradiated FDFMA3N109 MOSFETs. (Right:) Position of the irradiated MOSFETs in the flux 2D slice map of the reactor cavity.